

Figure 5: VM settings



Figure 6: Attaching an ISO to the VM's IDE controller

Configuring VM settings

When the VM is not running (i.e., when the guest OS is shut down, the VM is 'powered off'), the *Settings* toolbar button is enabled. The settings window enables you to fine-tune many VM settings. The network settings are visible in Figure 5. I've configured one virtual adapter, *Adapter 1*, attached to NAT (Network Address Translation). This option makes VirtualBox take network requests from the guest OS, send them out with the IP address of the host OS, and channel responses back to the VM over a private virtual network. To give the VM its own IP address on the LAN your host OS is connected to, you need to bridge the virtual adapter to the physical adapter on your host OS (which is usually *eth0* or *eth1*, but could also be *wlan0* if you're connected over Wi-Fi). You also need to ensure that there's a DHCP server active on the LAN, or otherwise allocate a proper static IP address in the network configuration of the guest OS.

Sharing data between the host and guest OS

You may have a large set of data accessible to your host OS, which you also want to access and use from the guest OS. How do you share the data with the guest OS? VirtualBox has a feature called *Shared Folders*, which lets you choose a folder (in the host OS) that is made available to the guest OS in the VM.

Installing guest additions

To use the *Shared Folders* feature, you have to install the

```
[root@bankur VBoxADDITIONS 3.1.6 59338]# ls
32bit      VBoxLinuxAdditions-amd64.run  VBoxWindowsAdditions.exe
64bit      VBoxLinuxAdditions-x86.run     VBoxWindowsAdditions-x86.exe
AUTORUN.INF VBoxSolarisAdditions.pkg
autorun.sh VBoxWindowsAdditions-amd64.exe
[root@bankur VBoxADDITIONS 3.1.6 59338]# ./VBoxLinuxAdditions-x86.run
Verifying archive integrity... All good.
Uncompressing VirtualBox 3.1.6 Guest Additions for Linux.....
VirtualBox Guest Additions installer
Building the VirtualBox Guest Additions kernel modules
Building the main Guest Additions module
Building the shared folder support module [ OK ]
```

Figure 7: GA installation in progress

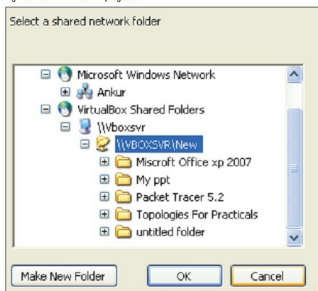


Figure 8: Browsing for the VirtualBox shared folder

VirtualBox *Guest Additions* (GA) into the VM: briefly, connect the GA ISO image as a CD media in the VM, and run the appropriate installer for your guest OS. (You only need to install GA once, not every time you want to connect a shared folder!) To do this, click the *Devices>Install Guest Additions* menu option on the VirtualBox window for your running VM. If the ISO image is not on your system, the OSE tries to download the appropriate ISO for your VirtualBox version from the website.

To manually download it, visit <http://download.virtualbox.org/virtualbox/> and choose the folder for your version of VirtualBox. Look for an ISO file whose name begins with 'VboxGuestAdditions'. If you copy or move the downloaded ISO to the */usr/share/virtualbox* directory on your system, VirtualBox should be able to find and automatically connect it when you click *Devices>Install Guest Additions*. Otherwise, you can also go to the *Storage* settings for your VM, connect the downloaded ISO image as a read-only media on the IDE controller (see Figure 6 for an example) and then start up the VM.

Depending on your guest OS, you may have to mount the GA ISO manually, though mostly it will be auto-mounted. Once it's mounted, the OS may auto-run the appropriate installer (Nautilus may offer a button to 'Open *Autorun Prompt*', which you can click). If that doesn't happen, open a terminal window as the root (use *su -* or *sudo -s* as appropriate for your guest OS). Change your working directory to the folder under */media* where the ISO is mounted (the folder name should begin